Epic 5: Dividing the Model into Train and Test Data

Introduction to Train and Test Data

Machine learning models require data for learning and evaluation.

The dataset is divided into training and testing sets.

Training data is used to teach the model.

Testing data evaluates model performance.

This process prevents biased evaluation.

The split ensures fair prediction results.

Most projects use an 80:20 ratio.

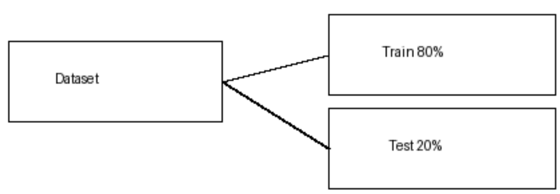
Sometimes 70:30 is also used.

The Human Development Index dataset follows the same concept.

Proper splitting improves generalization.

It reduces overfitting.

It is an essential ML step.



Training Dataset

Training data usually contains most of the records.

The algorithm learns patterns from this data.

Features and target values are provided.

Model parameters are adjusted during training.

Large training datasets improve learning.

Noise should be minimized before training.

Clean data produces better accuracy.

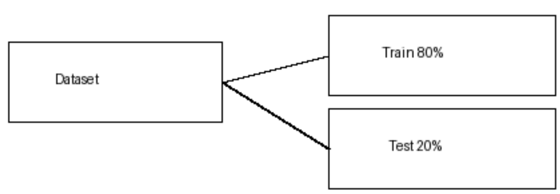
Training is repeated until learning is complete.

Cross-validation may also be used.

The trained model stores learned relationships.

It becomes ready for evaluation.

Training quality affects prediction performance.



Testing Dataset

Testing data is never used during training.

It measures the model's real performance.

Predictions are compared with actual values.

Metrics such as accuracy are calculated.

Testing identifies overfitting.

A good model performs well on unseen data.

Reliable testing increases confidence.

The test dataset should represent real-world data.

Results help improve the model.

Final evaluation is performed using testing data.

The trained model is validated.

This completes the train-test split process.